

Challenge (Habanero)

- Q1.** A recipe for making cookies requires $\frac{3}{4}$ cup of sugar. If you want to make half the recipe, how much sugar will you need in decimal form?
(A) 0.25
(B) 0.35
(C) 0.375
(D) 0.5
(E) 0.75
- Q2.** A building is 25.5 meters tall. If a model of the building is $\frac{1}{20}$ of the actual height, what is the height of the model in centimeters?
(A) 120
(B) 125
(C) 127.5
(D) 150
(E) 200
- Q3.** A cake recipe calls for $2\frac{1}{4}$ cups of flour. If you only have a $\frac{1}{8}$ cup measuring cup, how many times will you need to fill it to have enough flour?
(A) 16
(B) 17
(C) 18
(D) 20
(E) 36
- Q4.** A water tank can hold 4500 liters of water. If $\frac{3}{8}$ of the tank is already filled, how many more liters can be added in decimal form?
(A) 2925
(B) 3037.5
(C) 3150
(D) 3375
(E) 3825
- Q5.** A group of friends want to share some candy equally. If they have $4\frac{1}{2}$ bags of candy and there are 9 friends, how much candy will each friend get in decimal form?
(A) 0.4
(B) 0.5
(C) 0.6
(D) 0.75
(E) 0.9
- Q6.** A carpenter needs to cut a piece of wood into lengths of 2.5 meters. If the wood is 15.2 meters long, how many pieces can the carpenter cut?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9
- Q7.** A baker is making a cake that requires $\frac{1}{6}$ cup of oil. If the baker only has a $\frac{1}{8}$ cup measuring cup, how many times will the baker need to fill the measuring cup to have enough oil?
(A) 1.2
(B) 1.3
(C) 1.33
(D) 1.5
(E) 1.8
- Q8.** A water bottle can hold 1.8 liters of water. If $\frac{2}{5}$ of the bottle is already filled, how much more water can be added in decimal form?
(A) 1.2
(B) 1.4
(C) 1.44
(D) 1.6
(E) 1.8

Open Ended Questions (FRQ)

- Q9.** A construction project requires $15\frac{3}{4}$ tons of concrete. If the concrete is delivered in 2.5 ton trucks, how many trucks will be needed to deliver the concrete? Show your work and calculate the number of trucks required.
- Q10.** A chef is making a recipe that requires $3\frac{1}{2}$ cups of flour. If the chef only has a $\frac{1}{4}$ cup measuring cup, how many times will the chef need to fill the measuring cup to have enough flour? Additionally, if the recipe is being made for $\frac{3}{4}$ of the original amount, calculate the new amount of flour required in decimal form. Show your work and calculate the new amount of flour required.

Chef's Tips

- Ensure that students show their work and calculations for the free response questions.
- Remind students to convert between fractions, decimals, and percents where necessary.
- Encourage students to use the number line to compare and order rational numbers where applicable.